STAC WARS PS2

OPERATION DAYBREAK

Can the Rebellion’s base trust the Sun?

Dataset: SolarPrediction.csv | Rows analyzed: 32686 | Dates: 118

Solar radiation analysis, evidence-based inference, daily solar report, prediction, and relative solar-reliability novelty.

MISSION BRIEF

Headline finding 1 — Mean radiation peaks at 12:00 (727.474 dataset units). This is the best observed clock-hour average, not a guaranteed forecast.

Headline finding 2 — Strongest absolute Pearson relationship in the selected comparison is Temperature (r=0.735, n=32686). Correlation does not establish causation.

Headline finding 3 — Highest integrated daily radiation energy occurs on 2016-09-02 (7.957 kWh/m² under the W/m² assumption). This is radiation energy per area, not electrical output.

Recommendation for the base — treat days at or below the observed 20th-percentile daily energy (3.474 kWh/m²) as relative low-resource planning flags; preserve storage margin and verify against battery, load, and panel data before making operational decisions.

These are dataset-specific results. The idealised clear-sky envelope and day-type labels are heuristic proxies; they are not weather forecasts or engineering guarantees.

1. Dataset and Cleaning

UNIXTime int64

Data str

Time str

Radiation float64

Temperature int64

Pressure float64

Humidity int64

WindDirection(Degrees) float64

Speed float64

TimeSunRise str

TimeSunSet str

Datetime datetime64[us]

Date object

Hour int32

Minute int32

MinuteOfDay int32

DayOfYear int32

Month int32

HourSin float64

HourCos float64

DayOfYearSin float64

DayOfYearCos float64

SunriseDatetime datetime64[us]

SunsetDatetime datetime64[us]

MinutesSinceSunrise float64

DayLengthMinutes float64

DayLength_h float64

MinSinceSunrise float64

SolarFrac float64

IsDaytime bool

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"Temperature",

"Pressure",

"Humidity",

"WindDirection(Degrees)",

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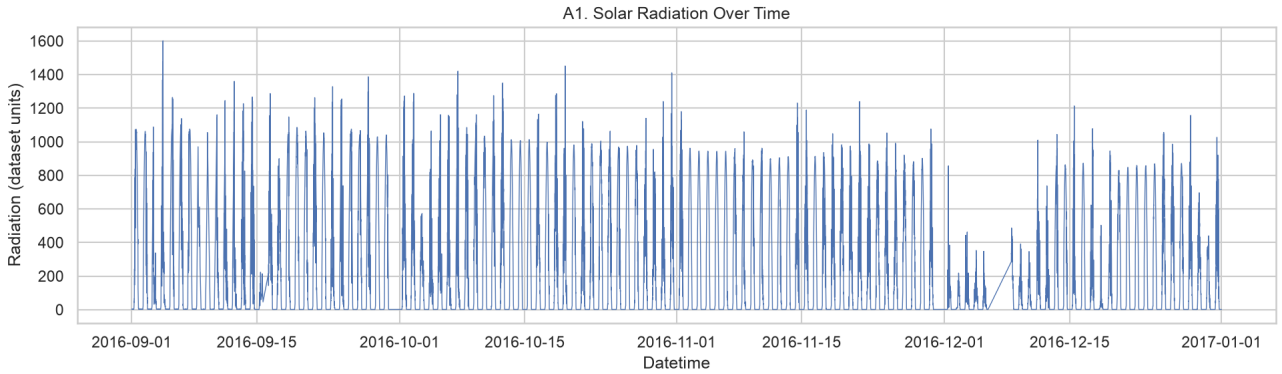
"Speed": 0,

Data quality audit table

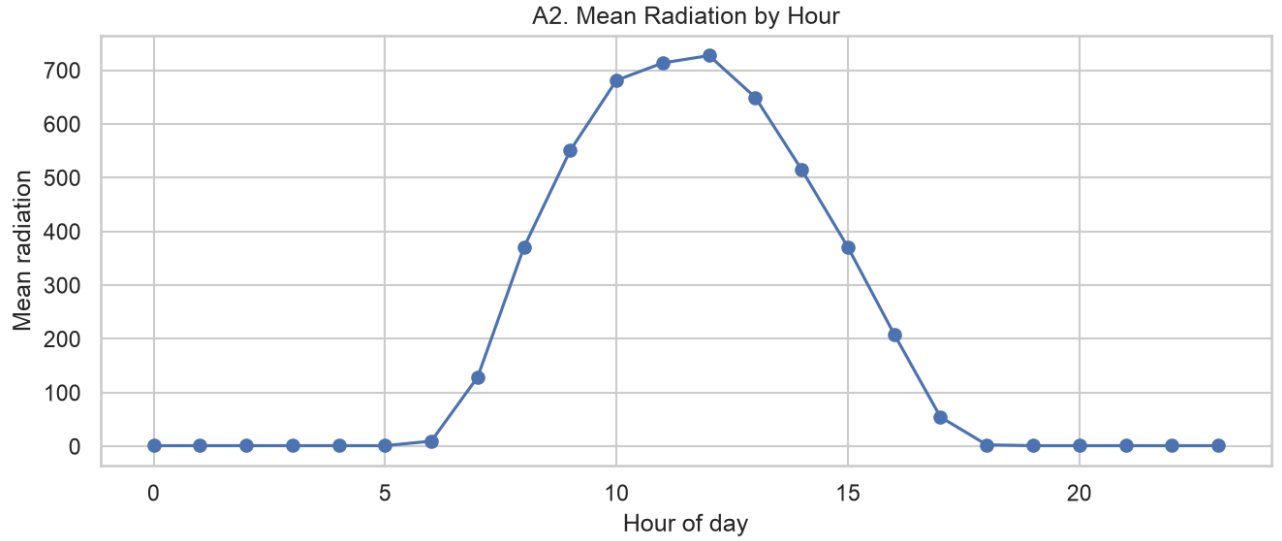
Issue	Count	Handling decision
Missing values	17447	Numeric gaps retained for audit; model pipelines impute features; target-missing rows excluded from modelling.
Negative radiation readings	0	Flagged for review; retained rather than silently clipping because they may be sensor/quality artefacts.
Duplicate timestamps	0	Reported; duplicate records retained in raw analysis. Consider aggregation if duplicates are confirmed accidental.
Large sampling gaps (>1.5× median interval)	809	Reported as gaps; no synthetic readings inserted.
Invalid timestamps removed	0	Rows without a parseable Datetime were removed before time-series analysis.

2. Part A — Visualizations

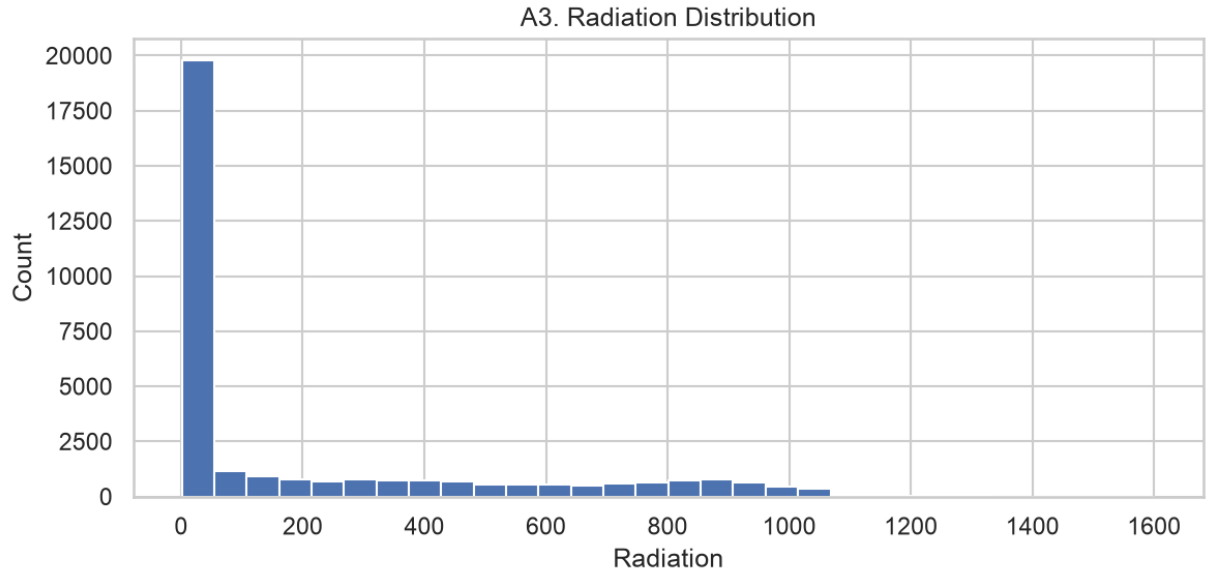
Radiation over time



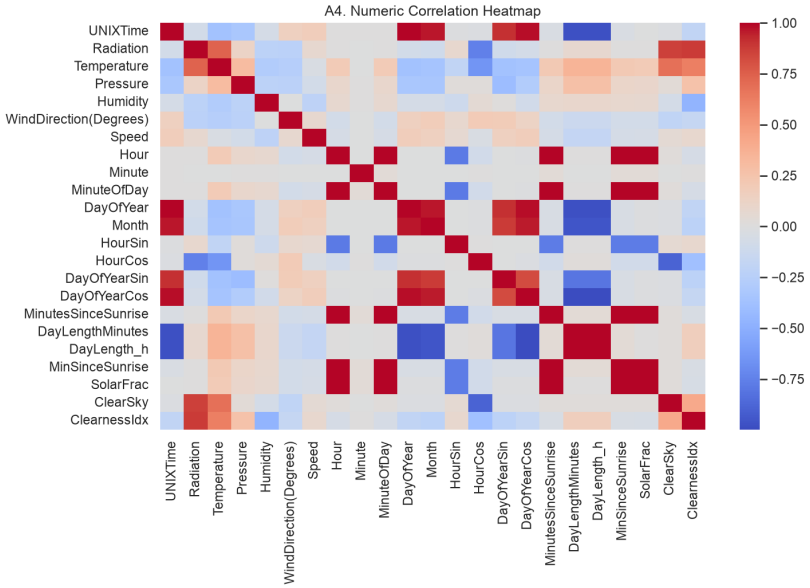
Mean radiation by hour



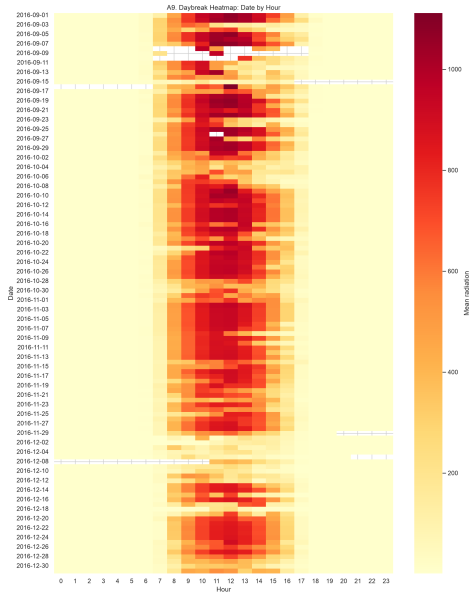
Radiation distribution



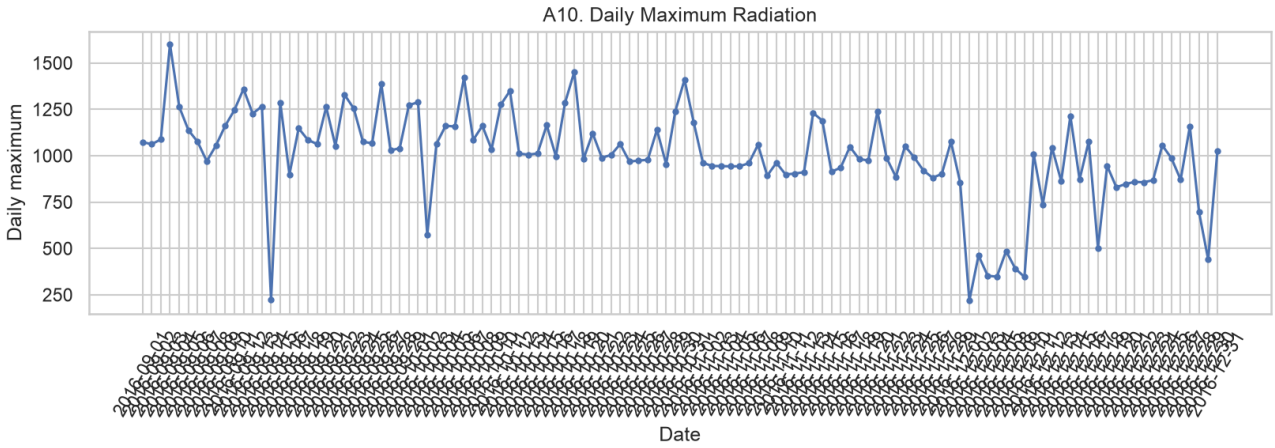
Numeric correlation heatmap



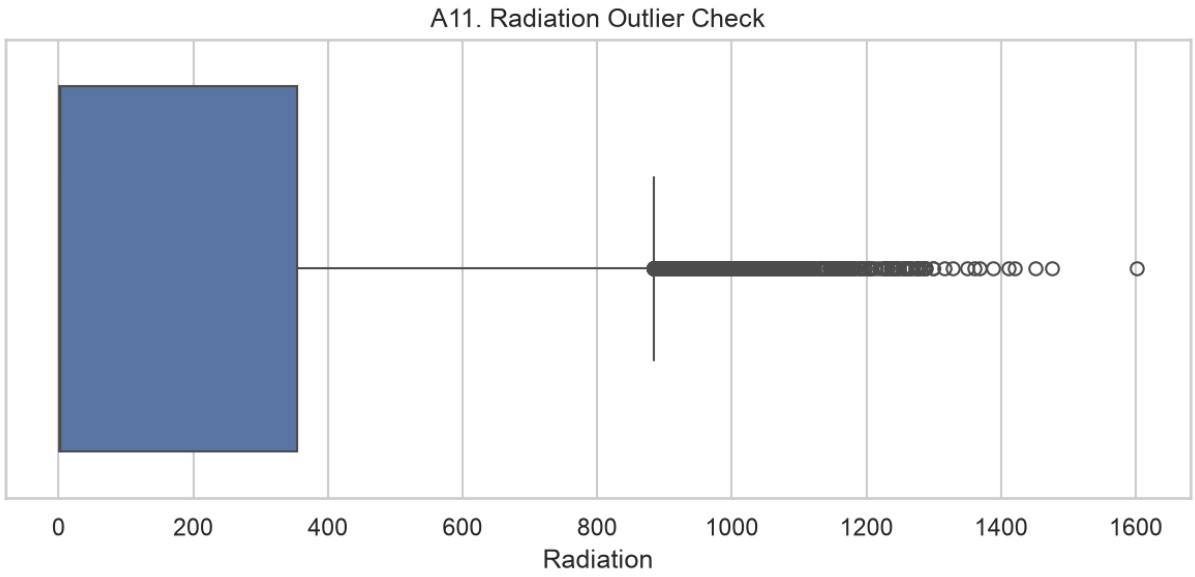
Date-hour heatmap



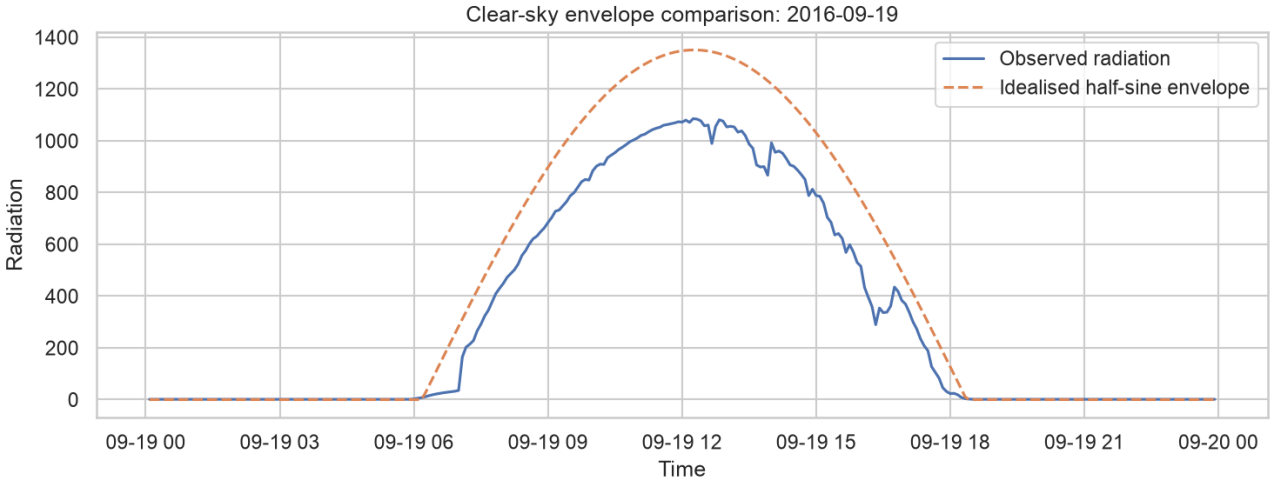
Daily maximum radiation



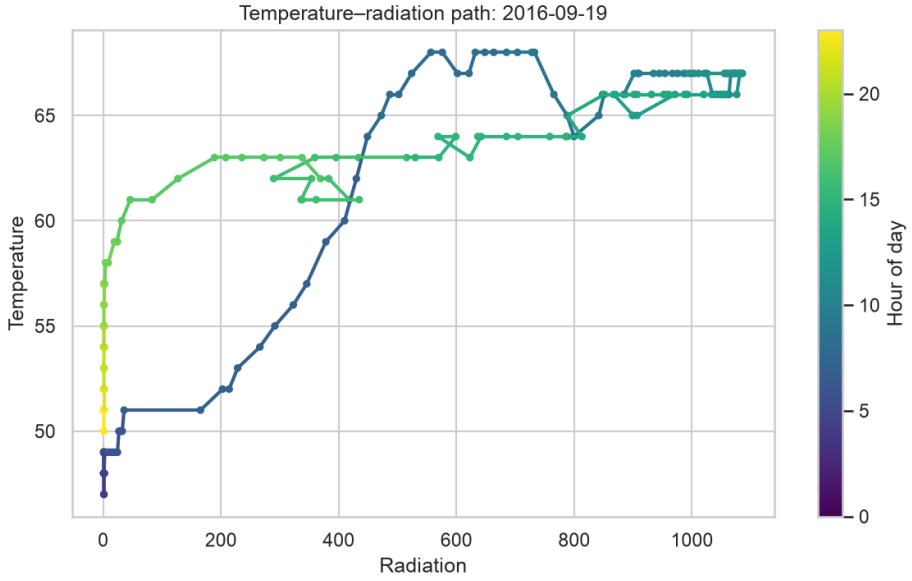
Radiation outlier check



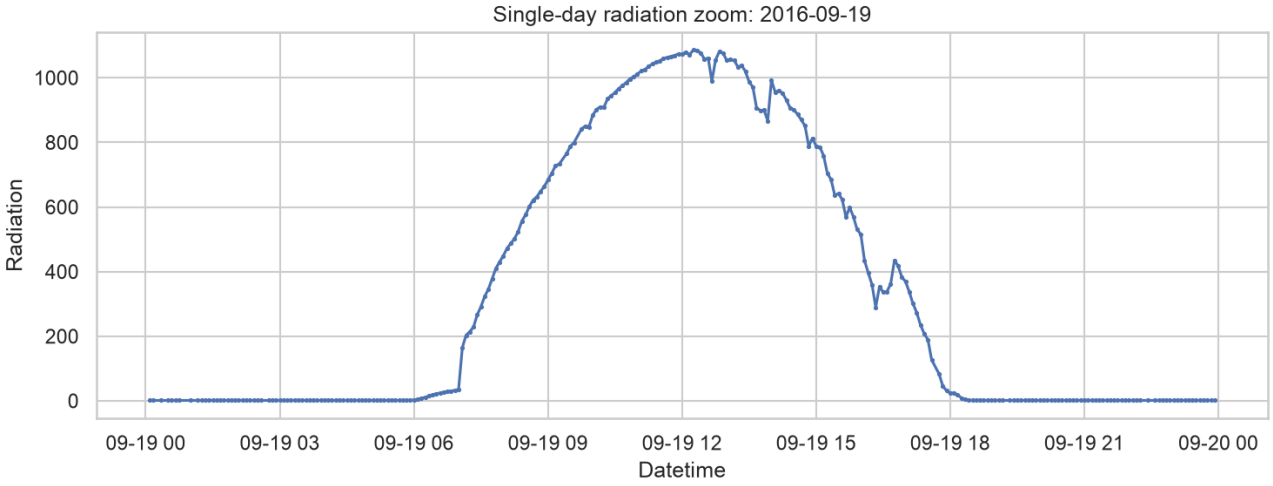
Idealised clear-sky envelope comparison



Temperature-radiation path coloured by hour



Single-day radiation zoom



3. Part B — Drawing Inferences

PART B — DRAWING INFERENCES

Evidence values are calculated from the supplied CSV. Correlation is not causation.

1. Highest mean radiation hour: 12:00, hourly mean 727.474. See graphs/A2_mean_radiation_by_hour.png.

2. Strongest absolute Pearson relationship (all_records): Temperature, r=0.735, Spearman rho=0.718, n=32686. Full ranking: data/radiation_parameter_correlations.csv.

2. Strongest absolute Pearson relationship (positive_radiation): Temperature, r=0.735, Spearman rho=0.718, n=32686. Full ranking: data/radiation_parameter_correlations.csv.

3. Radiation-temperature r=0.735; radiation-humidity r=-0.226; temperature-humidity r=-0.285. See the relevant scatterplots. Temperature may partly be a response to solar heating.

4. IQR outlier rule flags 1965 of 32686 non-missing radiation readings; fences=-528.277 to 883.742. Review data/radiation_iqr_outliers.csv for timestamps and context. Flags do not prove sensor failure.

Bonus thermal-lag check: hourly mean radiation peaks at 12:00 and temperature at 12:00; difference +0 hours. Verify against plots; aggregation and season coverage matter.

4. Part C — Solar Day Analysis

PART C — SOLAR DAY ANALYSIS

Datetime dtype: datetime64[us]

Daily max/mean and peak times: data/daily_report.csv

Top five days by average radiation:

1. 2016-09-19: mean=335.504, max=1085.750, n=275

2. 2016-09-02: mean=333.471, max=1062.010, n=279

3. 2016-09-22: mean=325.516, max=1053.170, n=284

4. 2016-09-25: mean=319.922, max=1075.290, n=284

5. 2016-09-28: mean=312.400, max=1029.300, n=282

Highest single reading: date=2016-09-04, time=12:15:04, radiation=1601.260.

Temperature=61.000; humidity=93.000; wind speed=9.000.

Sunrise=06:08:00, sunset=18:35:00, day length=12.45 hours.

Peak after sunrise=6 hours 7 minutes (367.1 minutes); daylight fraction elapsed=0.491; offset from solar-noon midpoint=-6.4 minutes.

5. Bonus — Prediction

BONUS — LEAK-AWARE SOLAR RADIATION PREDICTION

Random 80/20 split is the assignment baseline. Chronological 80/20 and GroupKFold by date reduce neighbouring-row/date leakage.

Temperature is partly caused by radiation and is suitable for nowcasting, not a clean forecast feature. Time-only is the forecast-style ablation.

Model ladder: idealised ClearSky baseline, Linear Regression, Random Forest, Gradient Boosting.

Partial dependence plot unavailable: Expected ax to have 3 axes, got 1

Quantile interval sample day: 2016-12-31; interval coverage on that sample day is computed in the CSV only after comparing actual readings with q10/q90; this interval is not guaranteed calibrated.

Results summary table (model × features × split × RMSE × R^2):

model features split test_RMSE test_R2

Idealised ClearSky ClearSky (proxy) clear_sky_baseline 354.263728 -0.257542

LinearRegression Time-only random_80_20 201.591921 0.592031

RandomForest Time-only random_80_20 77.338757 0.939955

GradientBoosting Time-only random_80_20 141.823115 0.798082

LinearRegression Time-only chronological_80_20 320.267997 -0.455301

RandomForest Time-only chronological_80_20 176.707123 0.556969

GradientBoosting Time-only chronological_80_20 215.975373 0.338189

LinearRegression Weather-only random_80_20 211.731284 0.549960

RandomForest Weather-only random_80_20 160.498838 0.741402

GradientBoosting Weather-only random_80_20 186.660707 0.650226

LinearRegression Weather-only chronological_80_20 203.280048 0.413706

RandomForest Weather-only chronological_80_20 179.416849 0.543278

GradientBoosting Weather-only chronological_80_20 186.076640 0.508742

LinearRegression Combined random_80_20 170.637634 0.707699

RandomForest Combined random_80_20 78.350355 0.938374

GradientBoosting Combined random_80_20 119.269009 0.857197

LinearRegression Combined chronological_80_20 170.172440 0.589130

RandomForest Combined chronological_80_20 125.663829 0.775949

GradientBoosting Combined chronological_80_20 130.835347 0.757129

LinearRegression Combined GroupKFold_by_date_5fold 169.202131 0.713133

RandomForest Combined GroupKFold_by_date_5fold 112.916481 0.872243

GradientBoosting Combined GroupKFold_by_date_5fold 121.368403 0.852402

Physics-informed RF (ClearnessIdx×ClearSky) Combined random_80_20 114.466642 0.880564

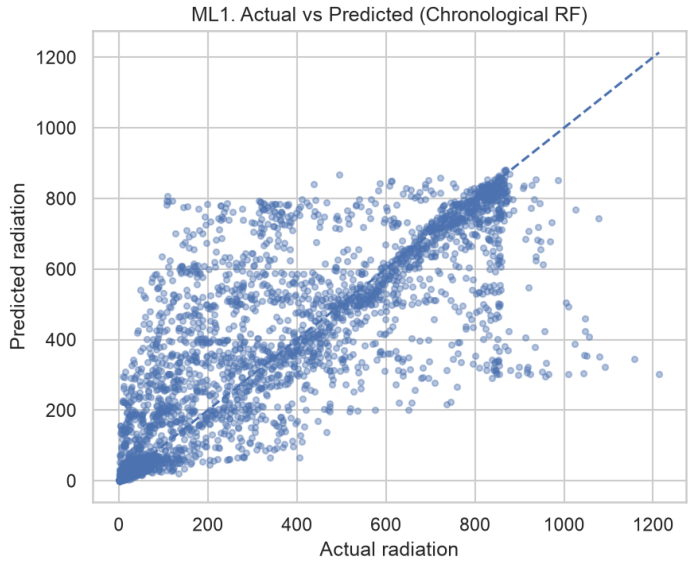
Physics-informed RF (ClearnessIdx×ClearSky) Combined chronological_80_20 181.284107 0.609169

Validation gap:

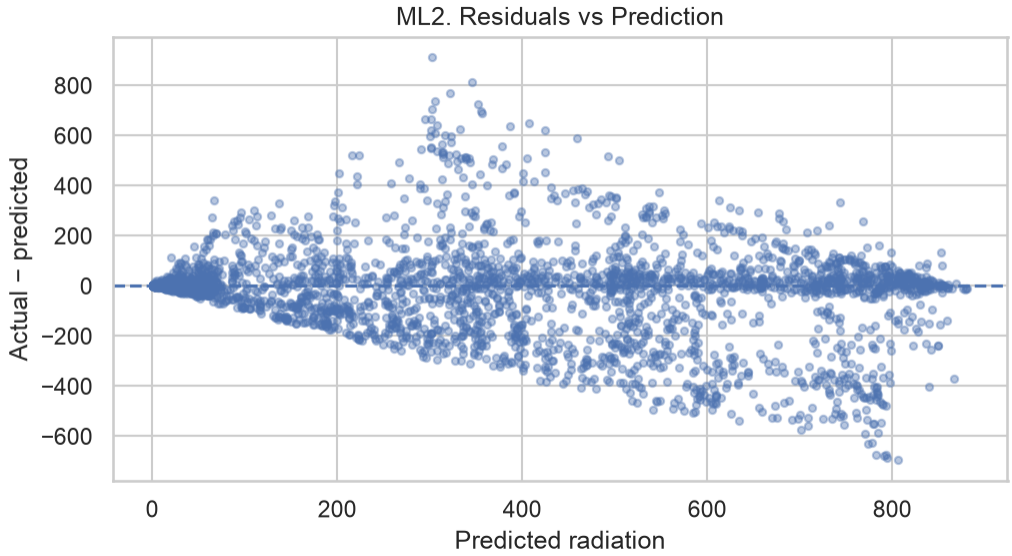
LinearRegression / Time-only: chronological RMSE minus random RMSE = +118.676; random RMSE=201.592, chronological RMSE=320.268.

LinearRegression / Weather-only: chronological RMSE minus random RMSE = -8.451; random RMSE=211.731, chronological RMSE=203.280.

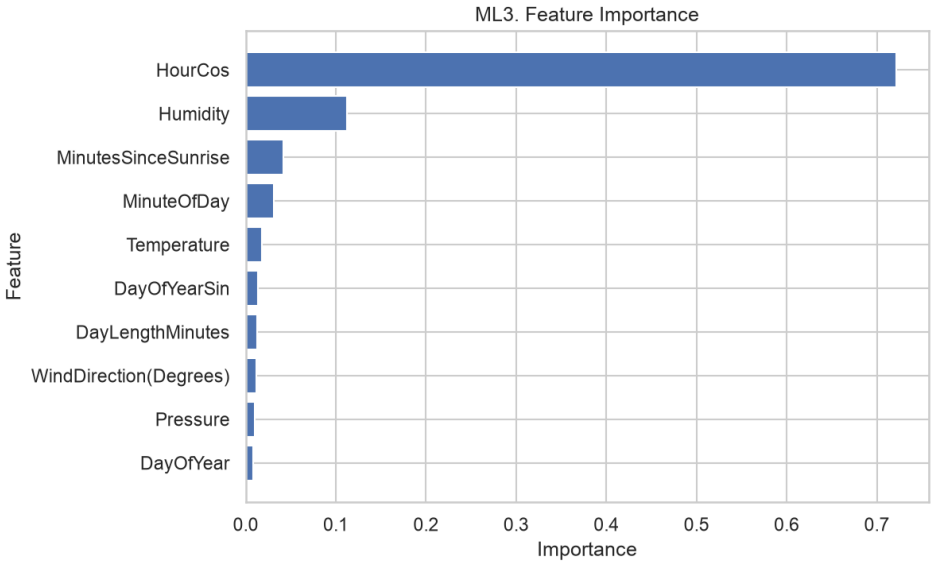
Actual vs predicted



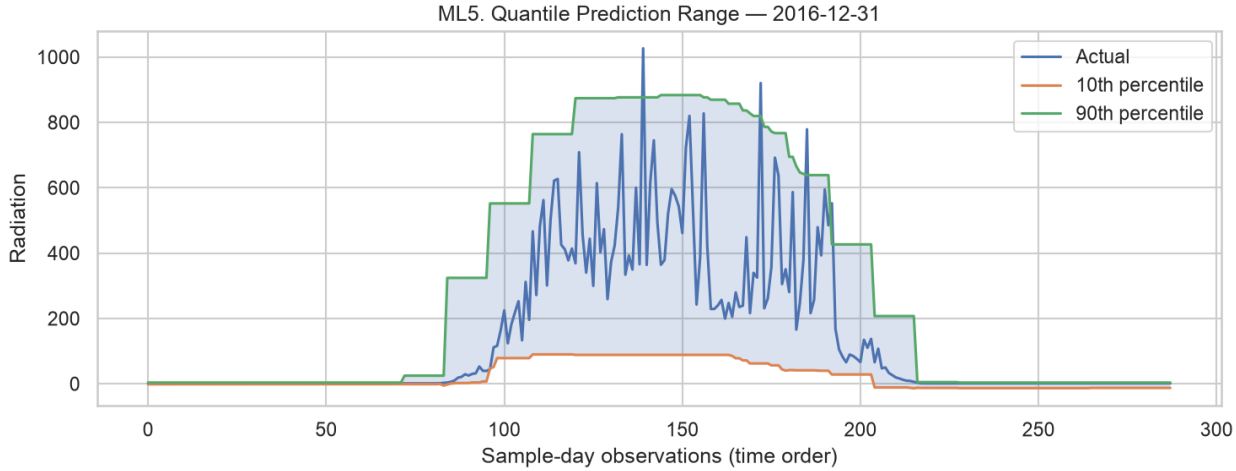
Residual diagnostics



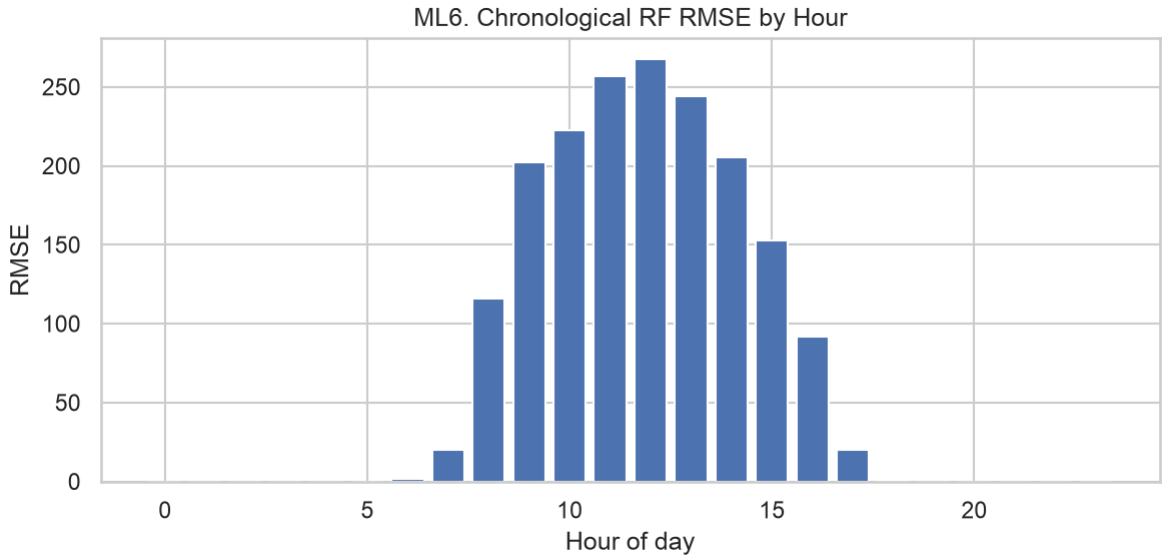
Random Forest feature importance



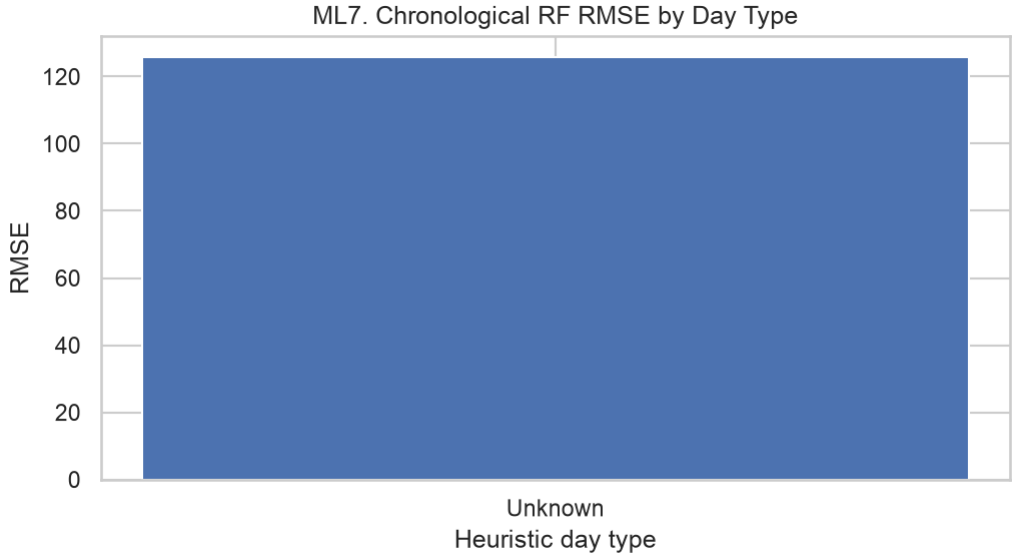
Quantile prediction interval for a sample day



Chronological test RMSE by hour



Chronological test RMSE by heuristic day type



6. Novelty — Operation Daybreak

OPERATION DAYBREAK NOVELTY

Solar Reliability Score is a relative ranking, not actual energy yield or blackout probability.

Low-radiation threshold: 1.200 (10th percentile of positive radiation).

Daylight is defined by sunrise/sunset when available; otherwise positive readings are used. Episodes split at gaps greater than 1.5 times median sampling interval.

Top reliability day: 2016-09-19, score 99.72/100, mean daylight radiation 643.966.

Longest low-radiation episode: 2016-12-01 06:50:03 to 2016-12-01 07:00:02, estimated 15.0 minutes.

Actual blackout risk requires panel capacity, battery state of charge, and load data.

7. Data Quality Audit

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ADDITIONAL PART B EVIDENCE

Hourly radiation peak is 12:00; mean offset from solar noon among readings in that hour is +15.1 minutes.

Hourly cross-correlation is highest at lag +1 hour(s) (correlation=0.939); this is an exploratory diurnal pattern, not proof of causal delay.

Humidity–clearness-index Pearson r=-0.470 (n=15239); clearness index is only a proxy because the clear-sky envelope is idealised.

Top mutual-information feature for all_records: ClearnessIdx, MI=1.215.

Top mutual-information feature for daytime_only: ClearnessIdx, MI=1.215.

8. Rebellion Daily Briefing and Base Power Planning

BASE POWER PLANNING — OPERATIONAL INDICATORS

Low-energy threshold is the 20th percentile of observed daily integrated radiation energy: 3.4741 kWh/m² (assuming radiation is W/m² and timestamps are representative).

Low-energy days: 24 of 118 days. This is a relative risk flag, not blackout probability.

Longest consecutive low-energy cluster: {'start': '2016-12-01', 'end': '2016-12-05', 'consecutive_days': 5}

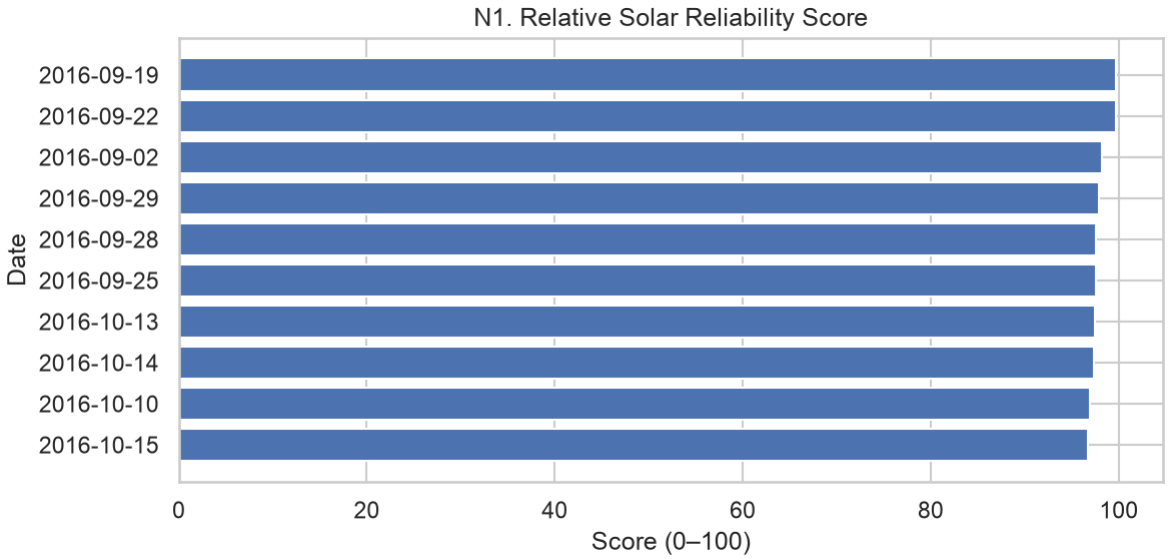
Day-type proportions: DayType PercentOfDays

Unknown 100.0

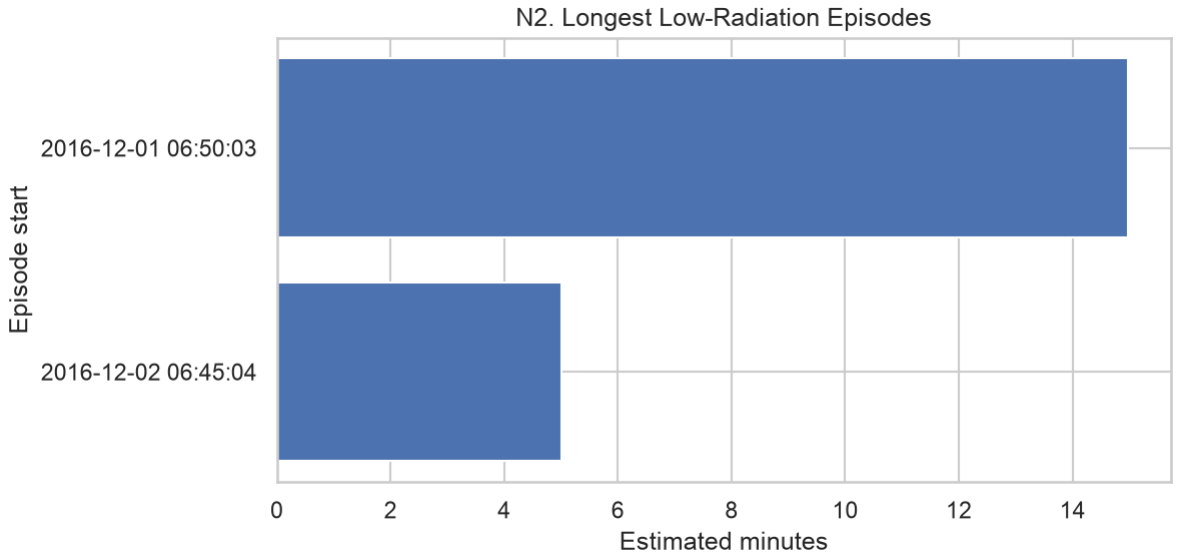
Hours most often at/above 80% of each day peak: 12:00 (57.2%), 11:00 (52.5%), 13:00 (50.1%)

ReadinessScore is min-max-normalized daily energy; it ranks days within this dataset and is not engineering-grade system sizing.

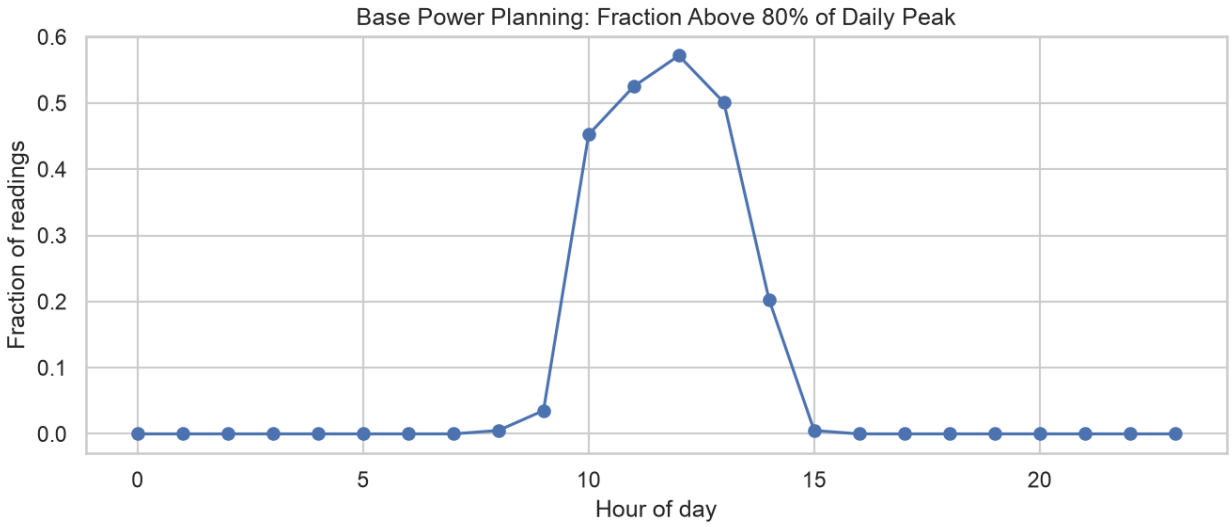
Relative reliability ranking



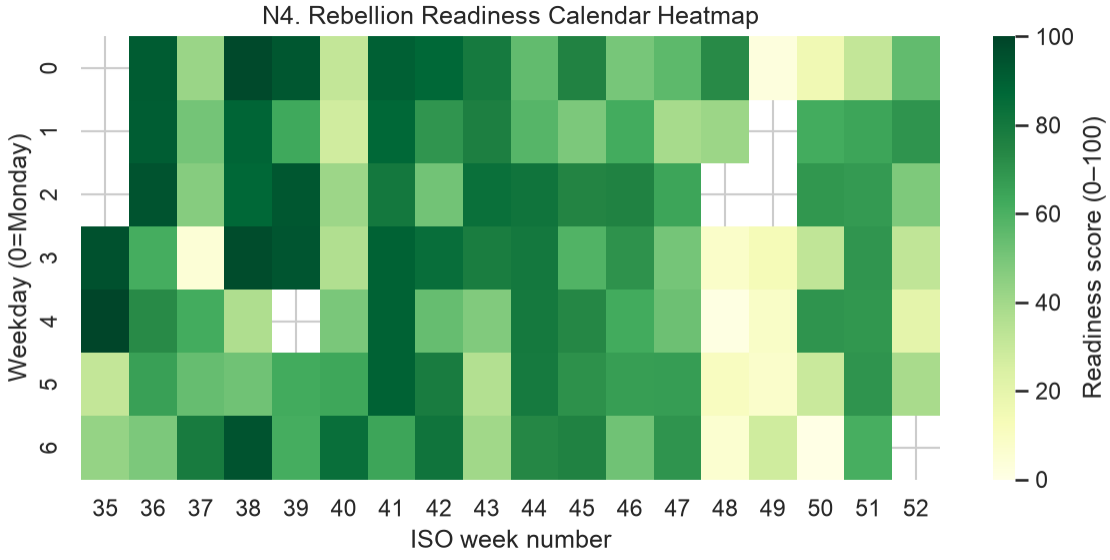
Sustained low-radiation episodes



Best operating window



Readiness calendar heatmap



9. Assumptions and Limitations

• Only the dates and season represented in the supplied file are analysed; one season or a short date range may not generalise to other seasons or years.

• No cloud-cover variable is assumed to exist. Clear/Partly cloudy/Overcast labels are heuristic bins of an idealised clearness proxy, not verified meteorological labels.

• Sensor noise, timestamp errors, duplicated records, and missing intervals can affect summaries. Outliers are candidates for review, not automatic proof of sensor failure.

• Flare caveat: readings above the idealised half-sine envelope may be cloud-edge enhancement, sensor noise, timing mismatch, or envelope-model error. They are not automatically “solar flares.”

• DailyEnergy_kWh_m2 assumes Radiation is W/m² and uses trapezoidal integration over observed timestamps; irregular gaps can make this estimate incomplete. It is not panel electrical yield.

• Temperature may partly be caused by solar radiation. Weather-feature models are nowcasts if temperature is measured at prediction time; time-only features are closer to forecast-style prediction.

• Quantile intervals are model estimates and are not guaranteed to have calibrated 80% coverage. The clear-sky proxy is not a validated atmospheric clear-sky model.

• Blackout-risk flags are relative low-resource indicators only. Actual outage risk requires load demand, panel efficiency/capacity, battery state of charge, storage losses, and system constraints.

Reproducibility: Python 3.14.9; dependencies listed in /Users/aashi/SolarSense/solar_analysis_output/requirements.txt; this single Python script was syntax-checked and run as a script on a synthetic smoke-test dataset during development. “Restart & Run All” is a notebook workflow and is not applicable to this standalone .py file. Re-run this script on the original CSV to reproduce the actual report.

What this means for the base: use these indicators to prepare and prioritise monitoring, not to promise solar availability. Confirm the site’s electrical system and weather forecasts before committing critical loads.